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Practical No 12(Post Lab)

Aim: To develop a Java program that converts units of **length,temperature, and weight** based on user input. The program allows the user to enter a value, choose the type of conversion, and display the converted result.

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DATE OF PERFORMANCE	
DATE OF SUBMISSION	
TOTAL MARKS (Performance + Viva + Lab Manual Submission)	

AIM: To develop a Java program that converts units of **length, temperature, and weight** based on user input. The program allows the user to enter a value, choose the type of conversion, and display the converted result.

OBJECTIVES:

- To develop a Java program for converting different units such as length, temperature, and weight.
- To allow users to input values and select the desired type of conversion.
- To perform accurate calculations using appropriate conversion formulas.
- To display the converted result in a user-friendly format.
- To understand and apply basic programming concepts like functions, loops, and conditional statements.

THEORY:

The Unit Converter is a simple application developed using the Java programming language that allows users to convert values from one unit to another. It is based on predefined mathematical formulas used for converting units of **length, temperature, and weight**. For example, kilometers can be converted into miles, Celsius into Fahrenheit, and kilograms into pounds using standard conversion factors.

The program follows a **menu-driven approach**, where the user selects the type of conversion and enters the value to be converted. The system then processes the input using conditional statements such as `if-else` or `switch` and performs the required calculation through functions. The result is then displayed to the user in an easy-to-understand format.

This project demonstrates the practical implementation of important programming concepts such as **user input handling (Scanner class), functions, loops, and decision-making statements**. It also shows how real-world problems can be solved using simple logic and structured programming. The Unit Converter is useful in everyday life, especially in fields like science, engineering, and education where unit conversions are frequently required.

Additionally, the program can be extended to include more types of conversions such as time, speed, and area, making it more versatile and powerful. Enhancements like error handling for invalid inputs and adding a graphical user interface (GUI) can further improve user experience and make the application more interactive and user-friendly.

Overall, this project serves as a foundational example for beginners to understand how software applications are designed to solve practical problems. It helps in building problemsolving skills and provides a strong base for developing more advanced applications in Java and other programming languages.

CODING:

```
import java.util.Scanner;
public class UnitConverter {
// Length Conversion
   public static double kmToMiles(double km) {
      return km * 0.621371;
   }

   public static double metersToKm(double meters) {
      return meters / 1000;
   }

   // Temperature Conversion
   public static double celsiusToFahrenheit(double c) {
      return (c * 9/5) + 32;
   }

   public static double fahrenheitToCelsius(double f) {
      return (f - 32) * 5/9;
   }

   // Weight Conversion
   public static double kgToPounds(double kg) {
      return kg * 2.20462;
   }

   public static double poundsToKg(double pounds) {
      return pounds / 2.20462;
   }

   public static void main(String[] args) {
Scanner sc = new Scanner(System.in);

      System.out.println("===== UNIT CONVERTER =====");
      System.out.println("1. Length Conversion");
      System.out.println("2. Temperature Conversion");
      System.out.println("3. Weight Conversion");
System.out.print("Choose option: ");
      int choice = sc.nextInt();

      double value, result;
```

```
      switch(choice) {
case 1:
            System.out.println("\n1. Kilometers to Miles");
            System.out.println("2. Meters to Kilometers");
System.out.print("Choose conversion: ");              int
lenChoice = sc.nextInt();
System.out.print("Enter value: ");              value =
sc.nextDouble();

            if(lenChoice == 1) {
result = kmToMiles(value);
               System.out.println("Converted value: " + result + " miles");
            } else if(lenChoice == 2) {
result = metersToKm(value);
               System.out.println("Converted value: " + result + " km");
            } else {
               System.out.println("Invalid choice!");
            }
            break;
         case
2:
            System.out.println("\n1. Celsius to Fahrenheit");
            System.out.println("2. Fahrenheit to Celsius");
System.out.print("Choose conversion: ");
            int tempChoice = sc.nextInt();

            System.out.print("Enter value: ");
value = sc.nextDouble();

            if(tempChoice == 1) {
               result = celsiusToFahrenheit(value);
               System.out.println("Converted value: " + result + " °F");
            } else if(tempChoice == 2) {
               result = fahrenheitToCelsius(value);
               System.out.println("Converted value: " + result + " °C");
            } else {
               System.out.println("Invalid choice!");
            }
            break;
         case
3:
            System.out.println("\n1. Kilograms to Pounds");
            System.out.println("2. Pounds to Kilograms");
System.out.print("Choose conversion: ");              int
weightChoice = sc.nextInt();
```

```
            System.out.print("Enter value: ");
value = sc.nextDouble();

            if(weightChoice == 1) {
result = kgToPounds(value);
                System.out.println("Converted value: " + result + " pounds");
            } else if(weightChoice == 2) {
                result = poundsToKg(value);


  System.out.println("Converted value: " + result + " kg");
            } else {
                System.out.println("Invalid choice!");
            }
            break;

default:
            System.out.println("Invalid option!");
        }

        sc.close();
    }
}
```

OUTPUT:

```
===== UNIT CONVERTER =====

1. Length Conversion
2. Temperature Conversion
3. Weight Conversion
Choose option: 1

1. Kilometers to Miles
2. Meters to Kilometers
Choose conversion: 1
Enter value: 10

Converted value: 6.21371 miles
```

1. Length Conversion
2. Temperature Conversion
3. Weight Conversion
Choose option: 2

1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
Choose conversion: 1
Enter value: 25

Converted value: 77.0 °F

1. Length Conversion
2. Temperature Conversion
3. Weight Conversion
Choose option: 3

1. Kilograms to Pounds
2. Pounds to Kilograms
Choose conversion: 2
Enter value: 100

Converted value: 45.3592 kg

CONCLUSION:

The Unit Converter project successfully demonstrates the use of Java programming to perform real-world calculations. It allows users to convert different units such as length, temperature, and weight in a simple and efficient manner. The menu-driven interface makes the program easy to understand and use.

During the development of this project, important programming concepts such as functions, conditional statements, and user input handling were applied. The use of separate methods for each conversion improves code readability and modularity. This makes the program easier to maintain and extend in the future.

REFERENCE:

1. **Java: The Complete Reference**

A comprehensive book covering core Java concepts including classes, methods, and user input.

2. **Head First Java**
   Beginner-friendly guide to understand Java programming in an interactive way.
3. **Oracle Java Documentation**
   Official documentation for Java programming and standard libraries.
4. **Programiz Java Tutorials**
   Provides easy explanations and examples for Java basics and programs.
5. **GeeksforGeeks Java**
   Helpful for understanding logic building and coding examples.
6. **W3Schools Java Tutorial**
   Simple and quick reference for Java syntax and programs.
7. **OnlineGDB**
   Useful online platform to write, compile, and run Java programs.